

14.384: Time Series Analysis.

Problem Set 2

due Thursday, October 8, 2026 before 8 pm

Please, submit solutions to Problem 3 (required) and to any one theoretical problem out of Problems 1-2.

1. Suppose that (y_t, x_t) evolve as

$$y_t = \beta x_t + u_t$$

$$x_t = \rho x_{t-1} + e_t$$

$$u_t = \phi u_{t-1} + \varepsilon_t$$

where $|\rho| < 1$, $|\phi| < 1$ and $\{e_t\}$ and $\{\varepsilon_t\}$ are mutually independent *iid* sequences both with mean 0 and variance 1.

- (a) Let $g_t = x_t u_t$. Show that g_t can be written as $g_t = \gamma g_{t-1} + a_t$ where $\gamma = \rho\phi$ and a_t is serially uncorrelated.

- (b) Let $\hat{\beta}$ denote the OLS estimator of β . Show that $\sqrt{T}(\hat{\beta} - \beta) \Rightarrow N(0, V)$ and derive an expression for V .

2. *Consistent Selection Procedure.* Consider a selection problem between models

$$(U) \quad y_i = \alpha x_i + \beta w_i + e_i;$$

$$(R) \quad y_i = \alpha x_i + e_i;$$

- (i) Let $\hat{M} = U$ if and only if $\left| \frac{\sqrt{n}\hat{\beta}_U}{\sigma_\beta} \right| > c_n$. Show that if $c_n \rightarrow \infty$ and $\frac{c_n}{\sqrt{n}} \rightarrow 0$ as $n \rightarrow \infty$, then the selection procedure is consistent.
- (ii) Assume that one decides to do estimation in the following way:

$$\hat{\alpha} = \arg \min_{\alpha} \min_{\beta} \left(\sum_{i=1}^n (y_i - \alpha x_i - \beta w_i)^2 - \lambda_n \mathbb{I}\{\beta = 0\} \right).$$

Show that for some choice of punishment for a larger model, λ_n , the result is equivalent to two step procedure from the lecture. What λ_n should be for consistency of the first step?

Hint. For part (ii) you may notice that $\min_{\alpha} \min_{\beta} = \min_{\beta} \min_{\alpha}$. First, concentrate out α , then notice that minimization with respect of β can be a corner solution, and show that it is equivalent to selection procedure described above.

3. *Spectrum and filters.* This is your first empirical problem. Choose a software package you feel comfortable using (I would recommend MatLab). You may use any users-written codes you find on Internet. Always make sure that the code is doing what you think it is doing. Please, do not forget to cite whatever you are using.

- (i) Go to Economic Database (FRED II) maintained by the Federal Reserve Bank of St. Louis. Download a macro series of your choice, I suggest quarterly values of Real GDP for the US as an example.
- (ii) Define the growth rate for real GDP. Estimate and plot spectrum for the growth rate. Discuss the graph. Find which peak in the spectrum corresponds to business cycles.
- (iii) Use the following three cycle removing devices: a) run the OLS to detrend the series ; b) use Prescott-Hodrick filter; c) apply Baxter-King filter.
- (iv) Re-estimate spectrum for all series after applying each of the three procedures. Draw spectrum functions. Discuss the differences.

Note. As in real life empirical research, you will need to make a lot of choices while performing the task, such as choosing lag length, kernel function, and so on. Try to be reasonable, always check whether your results are sensitive to these choices. Also check original papers for suggestions.